

VWAttacker: A Systematic Security Testing Framework for Voice over WiFi User Equipments

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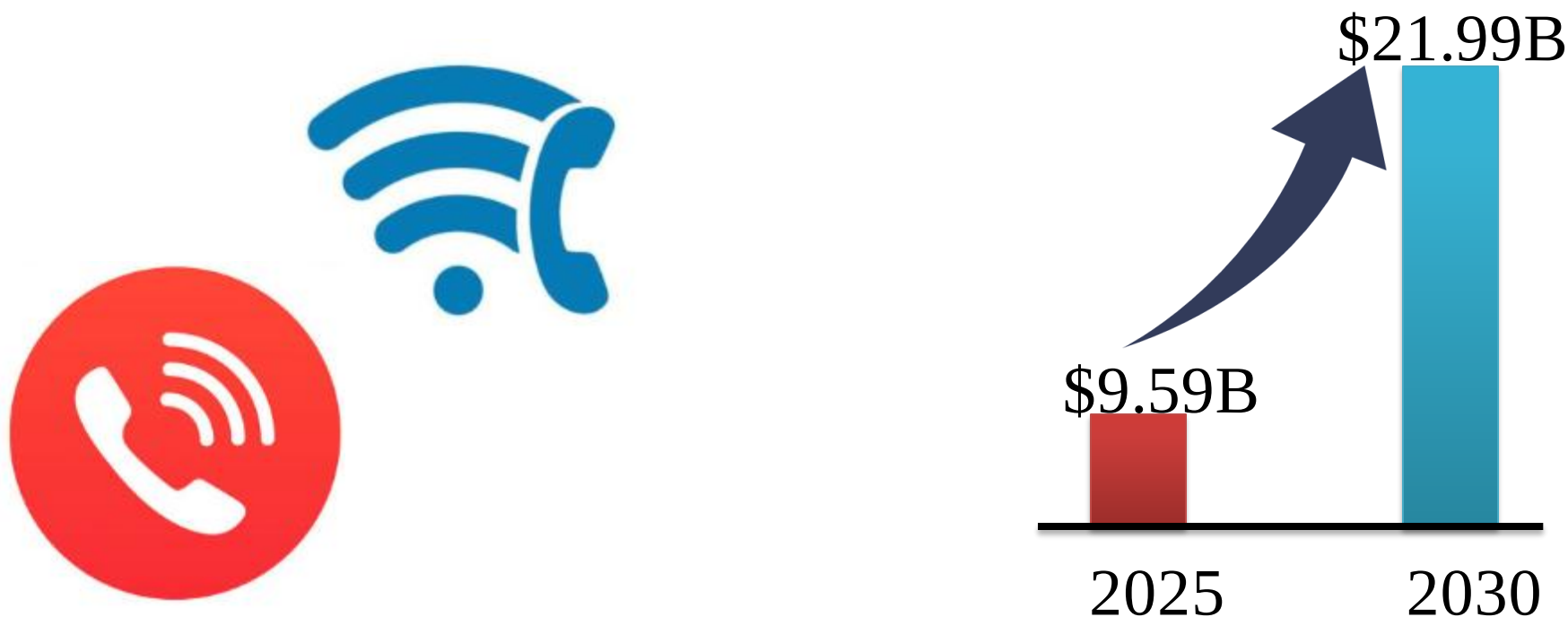
Voice over WiFi (VoWiFi)

VoWiFi is a mobile call service to extend network coverage by utilizing WiFi in areas where a cellular network is unavailable



Voice over WiFi (VoWiFi)

VoWiFi usage is increasing and its market is estimated to reach USD 21.99B in 2030 from USD 9.59B in 2025 with 18% of CAGR



Global VoWiFi Market Size

Security Analysis on VoWiFi

As the widespread use of VoWiFi requires robust security guarantees, there have been several works to analyze security of VoWiFi

Paper	Systematic Framework	Implementation Analysis	Complete VoWiFi UE Testbed	Scalable Property Extraction
Wi-Not-Calling [6]		✓		
Xie <i>et al.</i> [41]		✓		
Lu <i>et al.</i> [25]		✓		
Shi <i>et al.</i> [36]		✓		
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There is **no systematic framework** for analyzing VoWiFi implementations **with a complete open-source VoWiFi UE testbed**

Security Analysis on VoWiFi

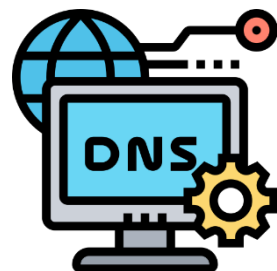
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VWAttacker	✓	✓	✓	✓

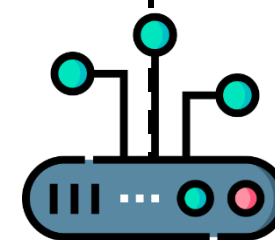
We develop the first **systematic** and **adversarial property-guided** framework to evaluate the security of VoWiFi UE implementations

VoWiFi Protocol

Internet Mobile Network



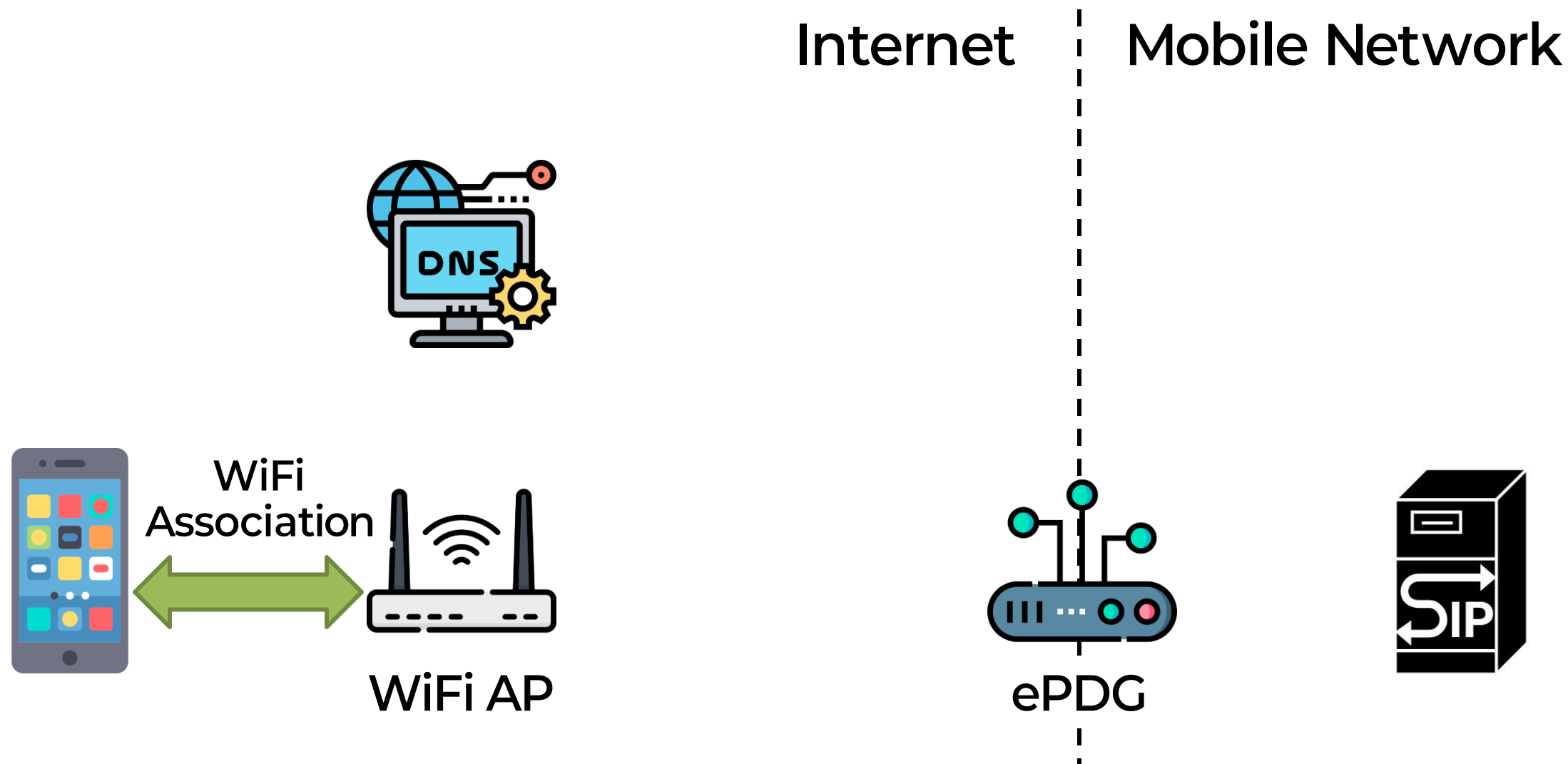
WiFi AP



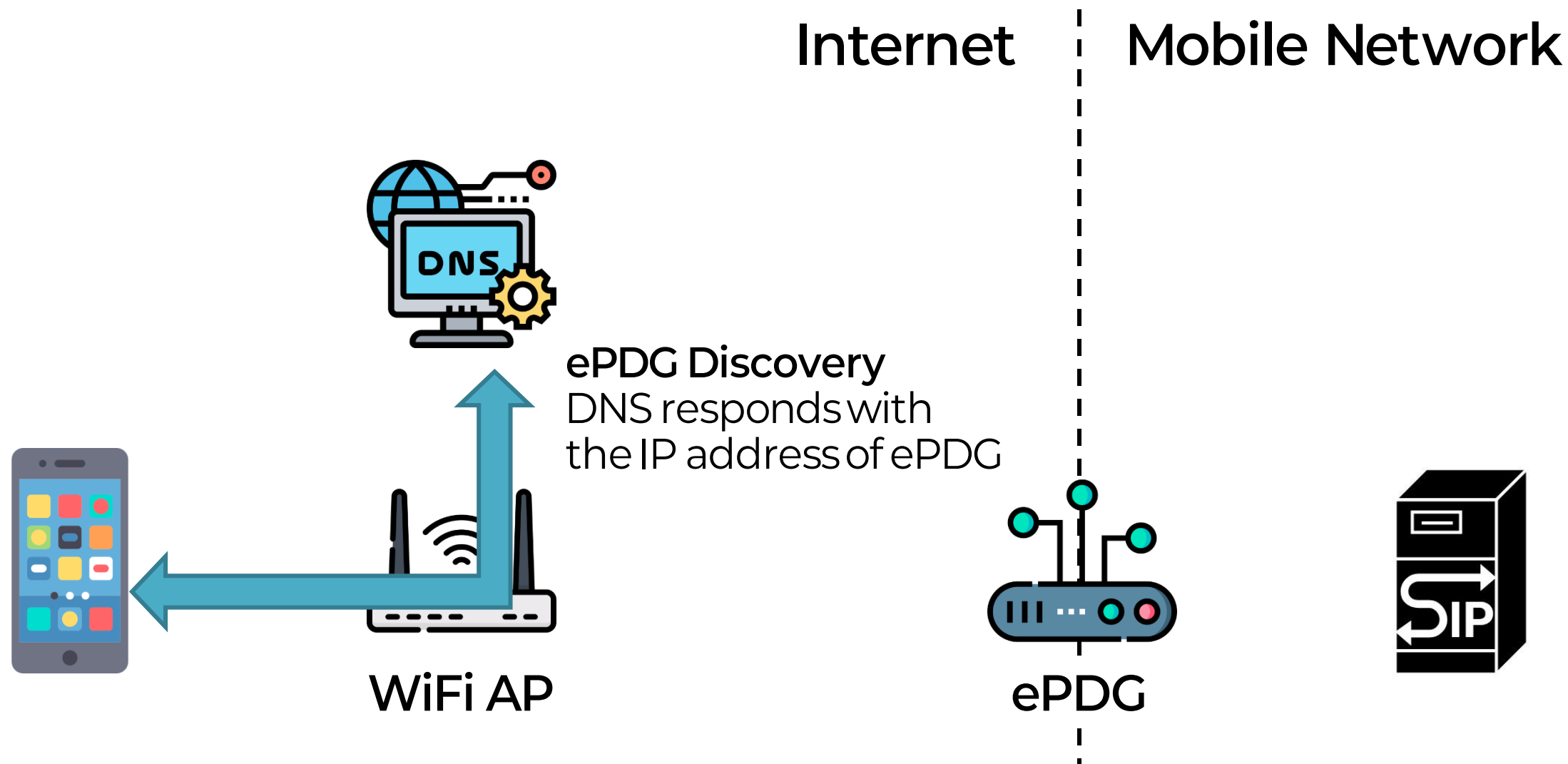
ePDG



VoWiFi Protocol



VoWiFi Protocol



VoWiFi Protocol

Internet

Mobile Network

UE Registration

UE establishes the VPN channel with ePDG

- IKEv2 and EAP-AKA



VoWiFi Protocol

Internet

Mobile Network

UE Registration

UE establishes the VPN channel with ePDG

- IKEv2 and EAP-AKA

Call

UE makes or receives a call through VPN

- IPSec and SIP



VWAttacker

An adversarial property-guided testing framework to evaluate the security of VoWiFi UEs, addressing the following three challenges:

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(C1) VoWiFi network testbed

No available open-source VoWiFi UE testbed for security testing

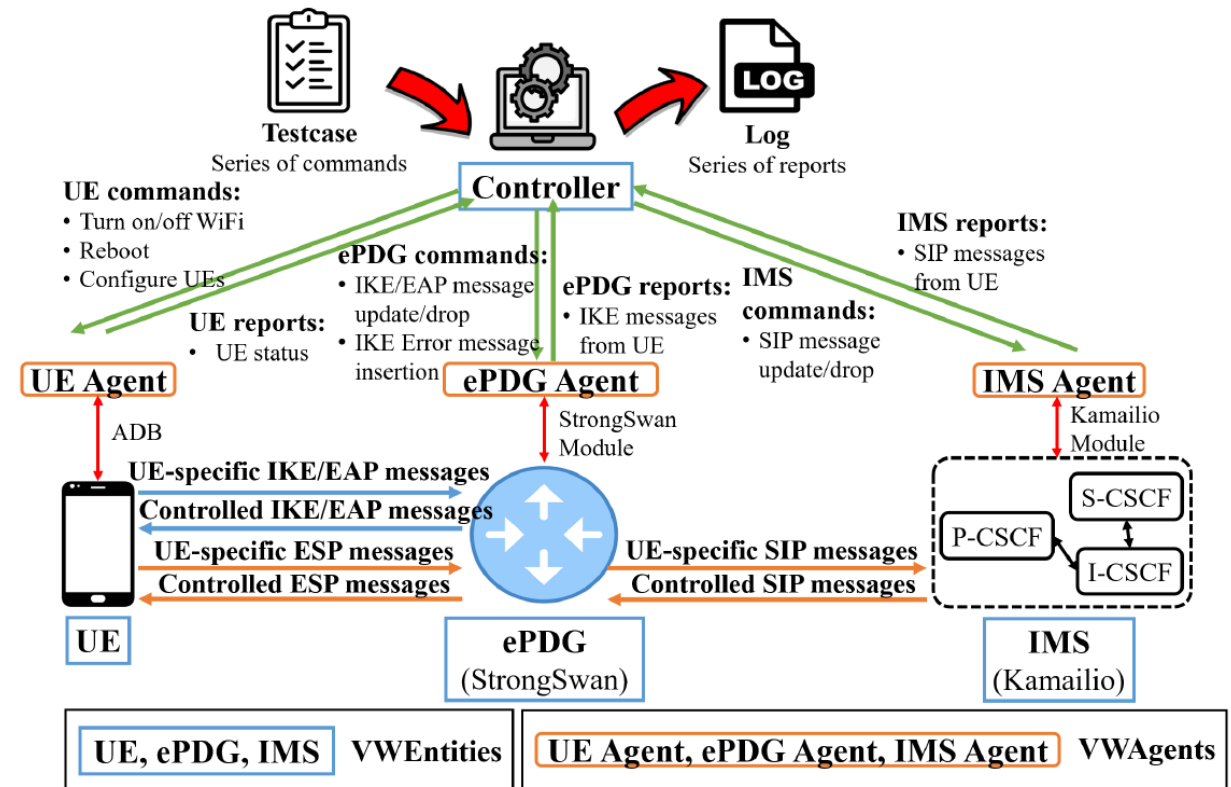
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→ We build a VoWiFi network testbed with StrongSwan, Kamailio, and a custom SIM card, which are all controllable



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(C2) Scalability of property-guided testing

(C2.1) Scalable and accurate property extraction

Manual extraction is time-consuming and labor-intensive

- LLM-based property extractor
- Property-to-testcase encoder

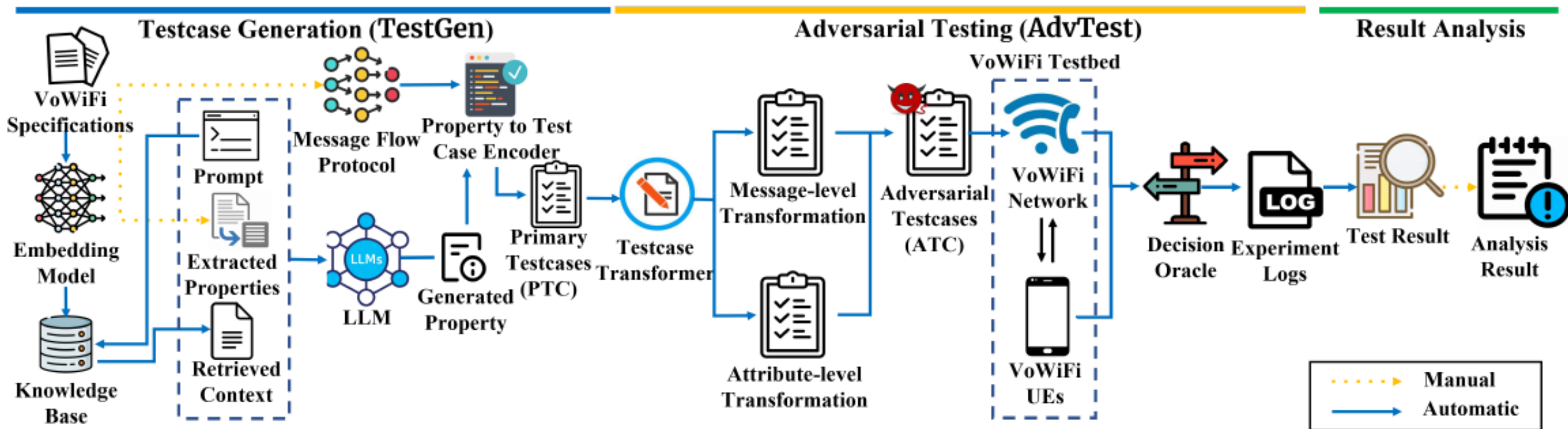
(C2.2) Property-violation detection

No formal canonical model to detect deviation or violation

- Two decision oracles

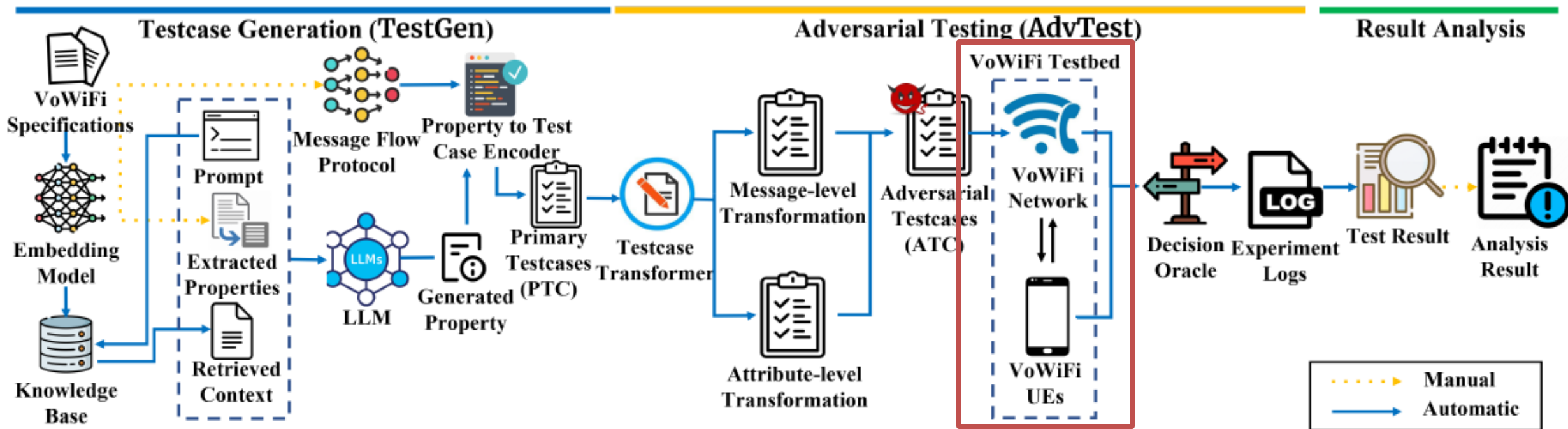
Overview

VWAttacker runs over a VoWiFi testbed



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VoWiFi Testbed

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(R3) Malleable testing system

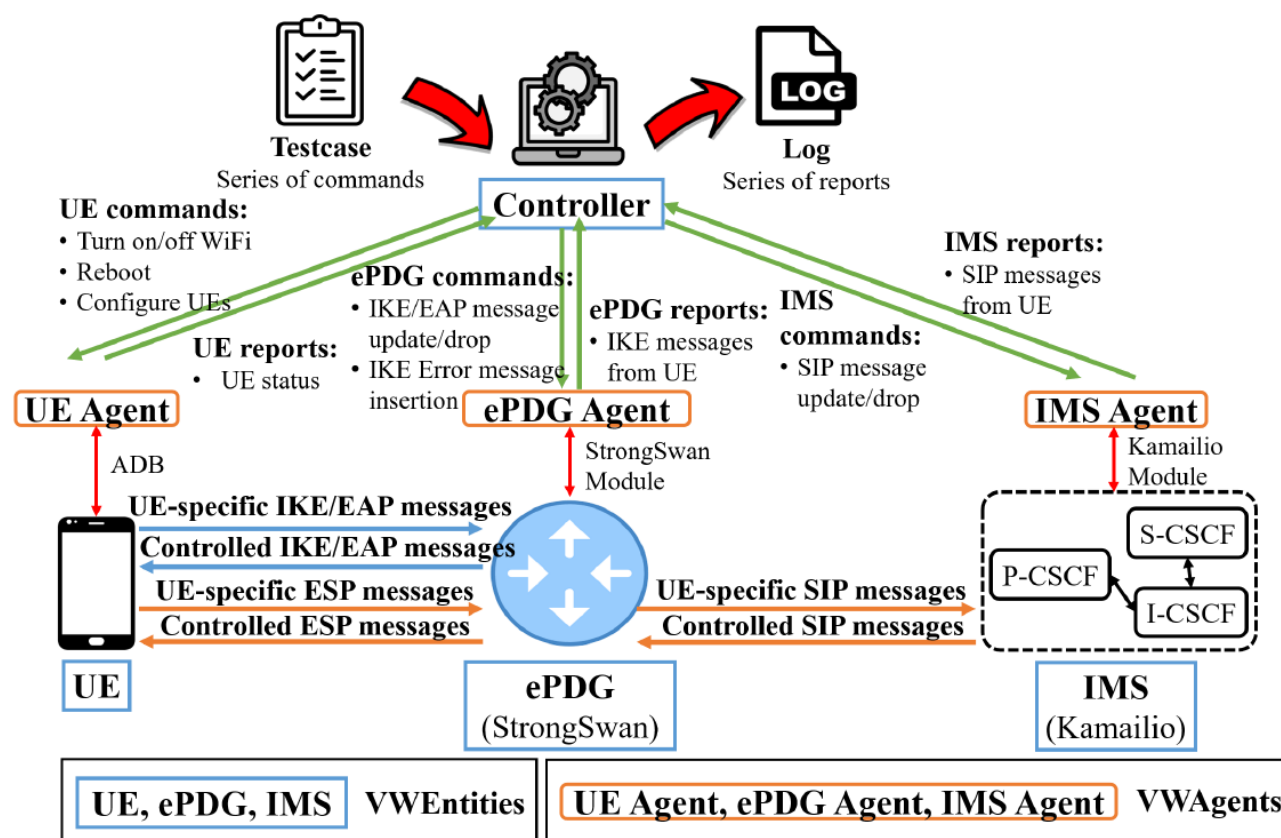
We should be able to send any test input to UEs

VoWiFi Testbed

The testbed provides a JSON-based interface for testers to describe testcases (TCs), and follows the central-peripheral paradigm

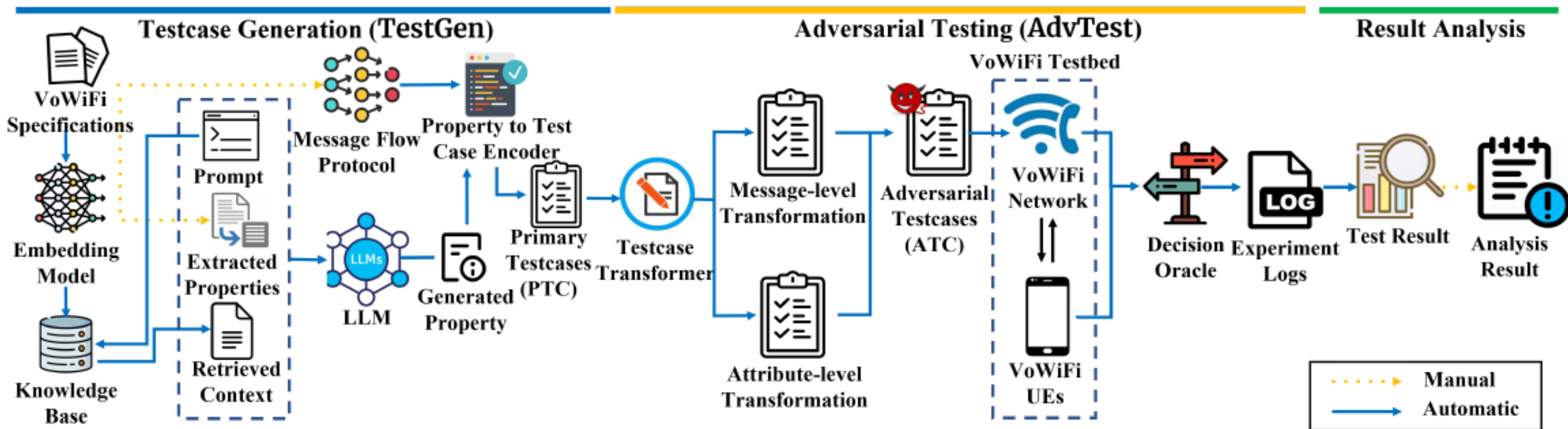
```

1 {"testcases": [
2   {"testcase": [
3     {"receiver": "ue",
4      "name": "enable_vowifi",
5      "reporter": "epdg",
6      "id": "turn-on-wifi"
7   }],
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12      "id": "init-enc-algo-13" },
13     {"receiver": "epdg",
14      "name": "ike_sa_init_response",
15      "reporter": "epdg",
16      "sub": [
17        {"name": "security_association",
18         "sub": [
19           {"name": "transform",
20            "sub": [
21              {"name": "encryption_algorithm",
22               "type": "uint16",
23               "value": 13,
24               "op": "update"
25            }
26          ]
27        }
28      ]
29     }
30   ]
31 }
  
```



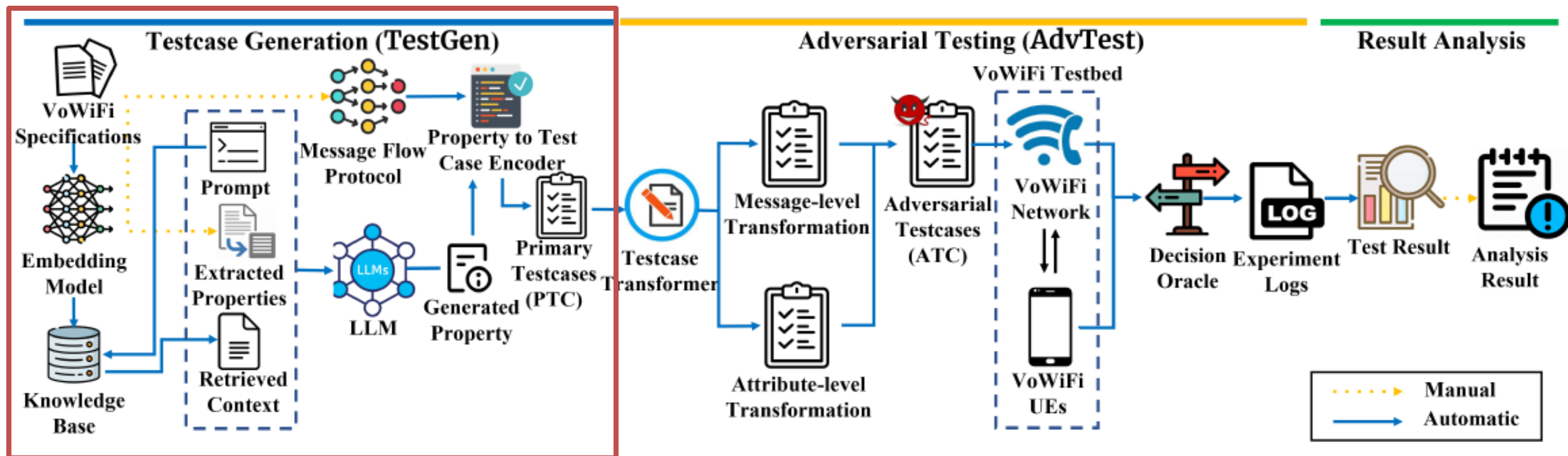
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VWAttacker consists of two core modules called TestGen and AdvTest



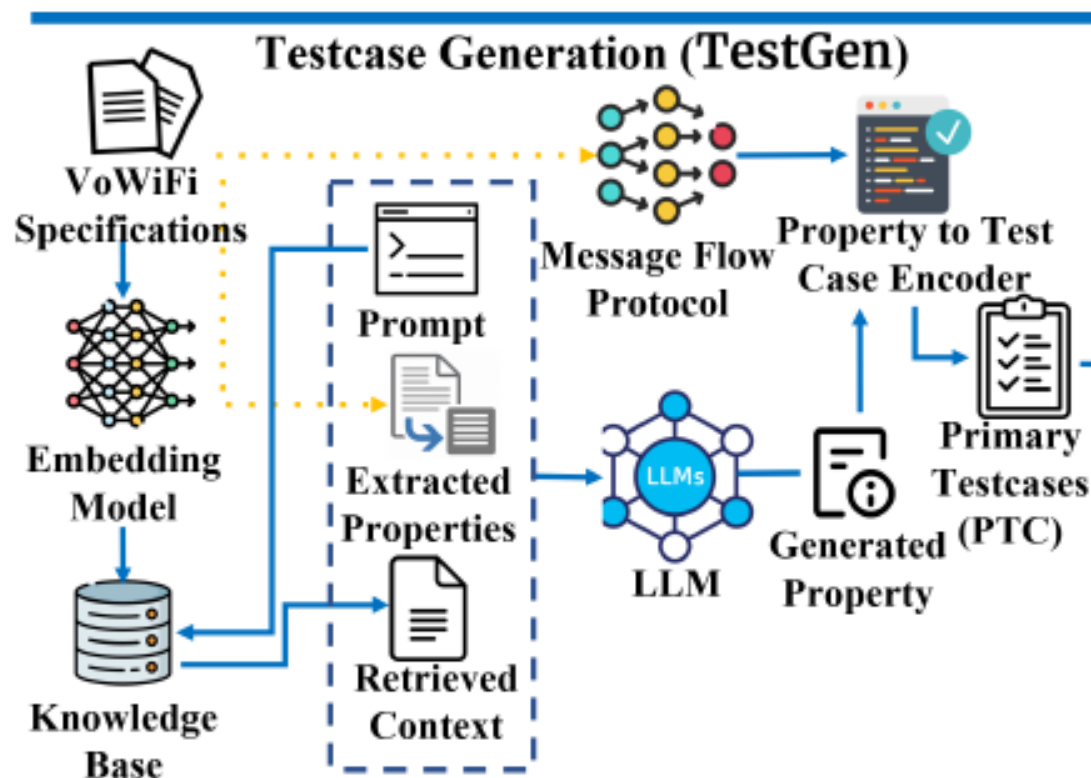
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It extracts properties from specifications using LLM and encodes them into primary testcases (PTCs)



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3GPP specifications
+ RFC documents

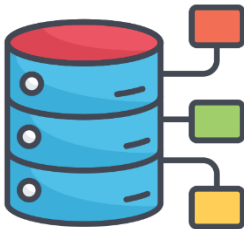
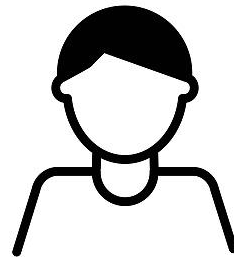


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Example properties
(requiring minimal effort)



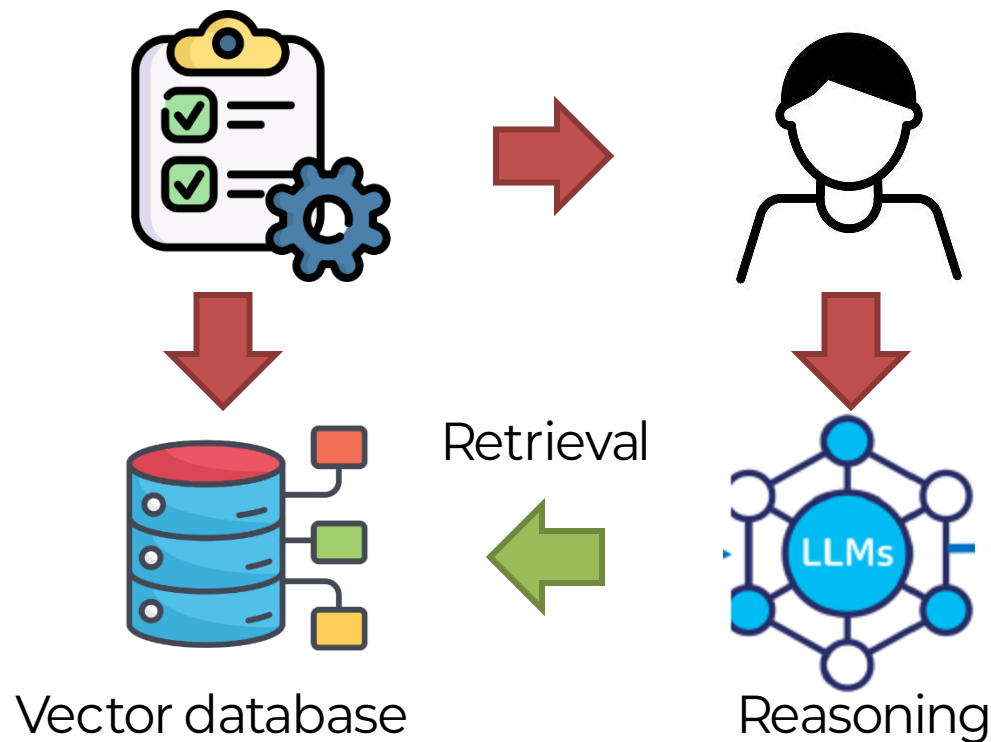
Vector database

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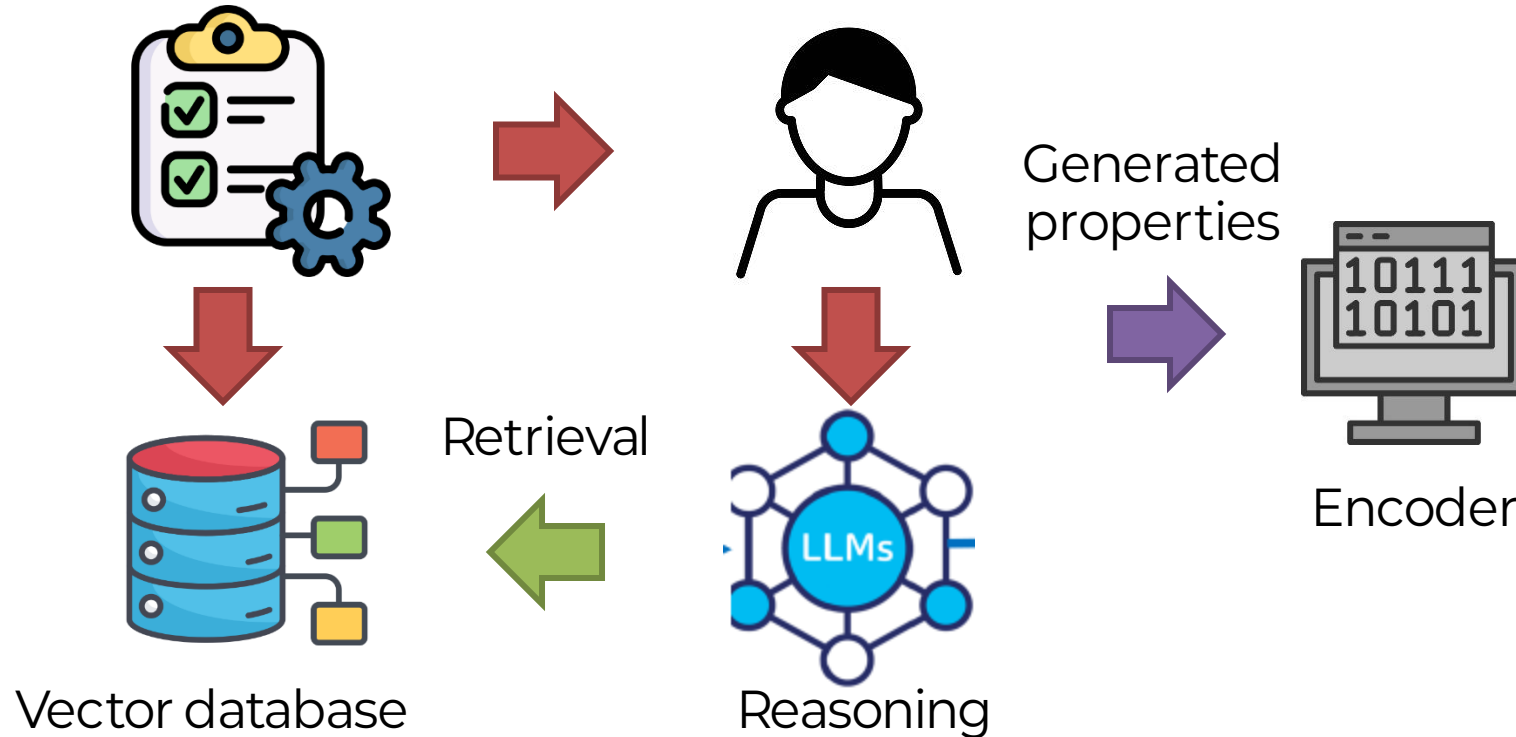


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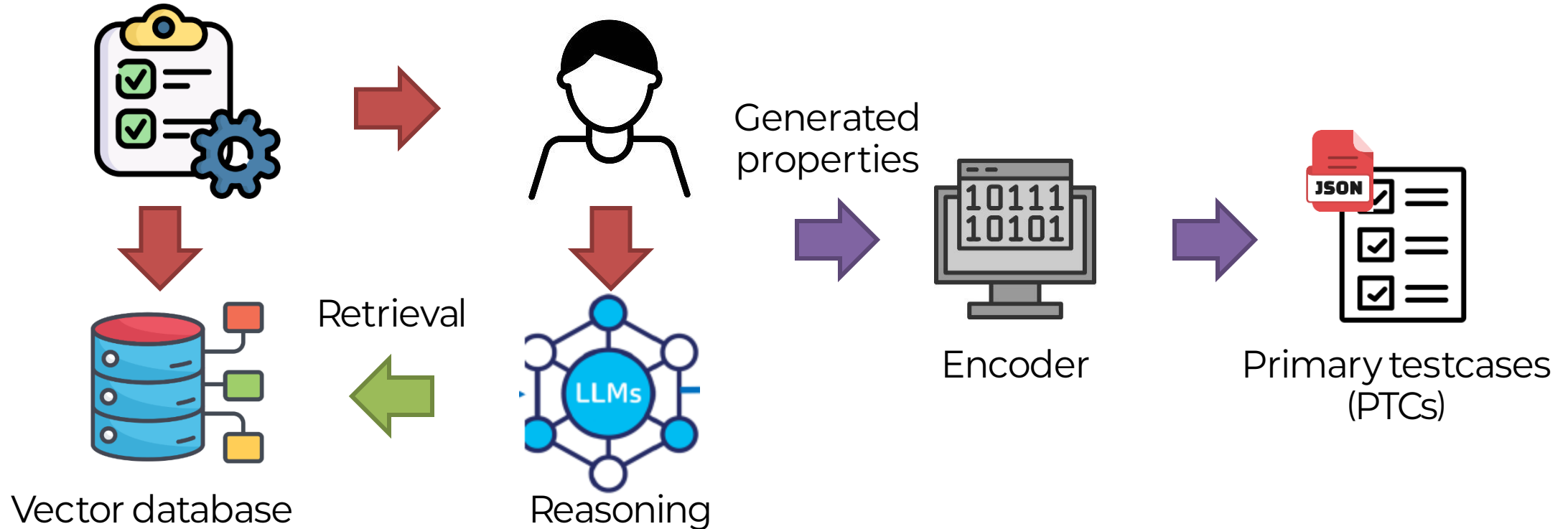


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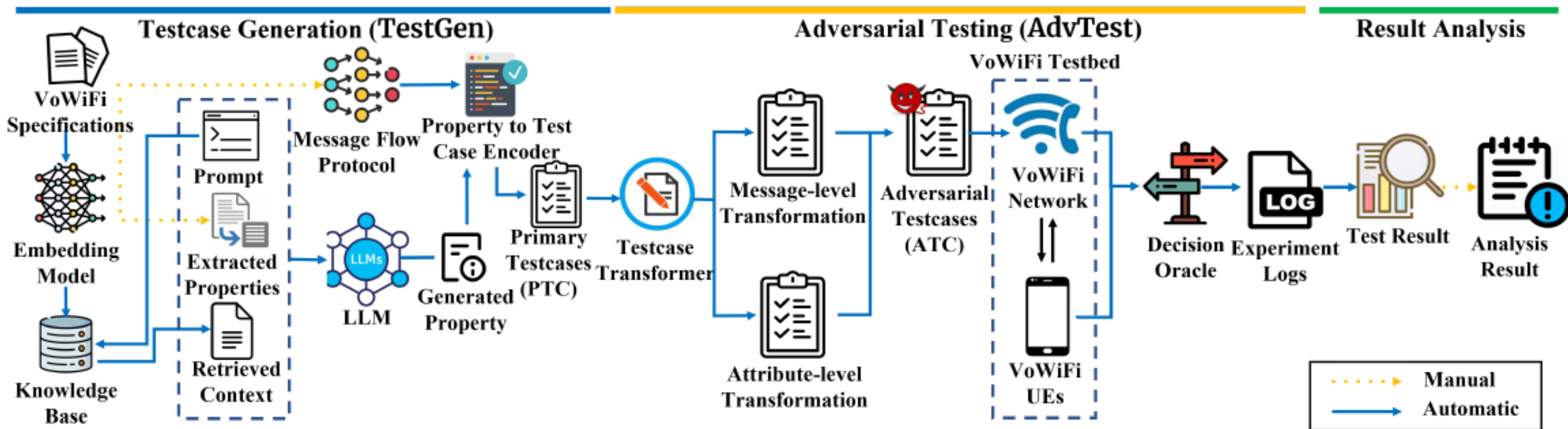
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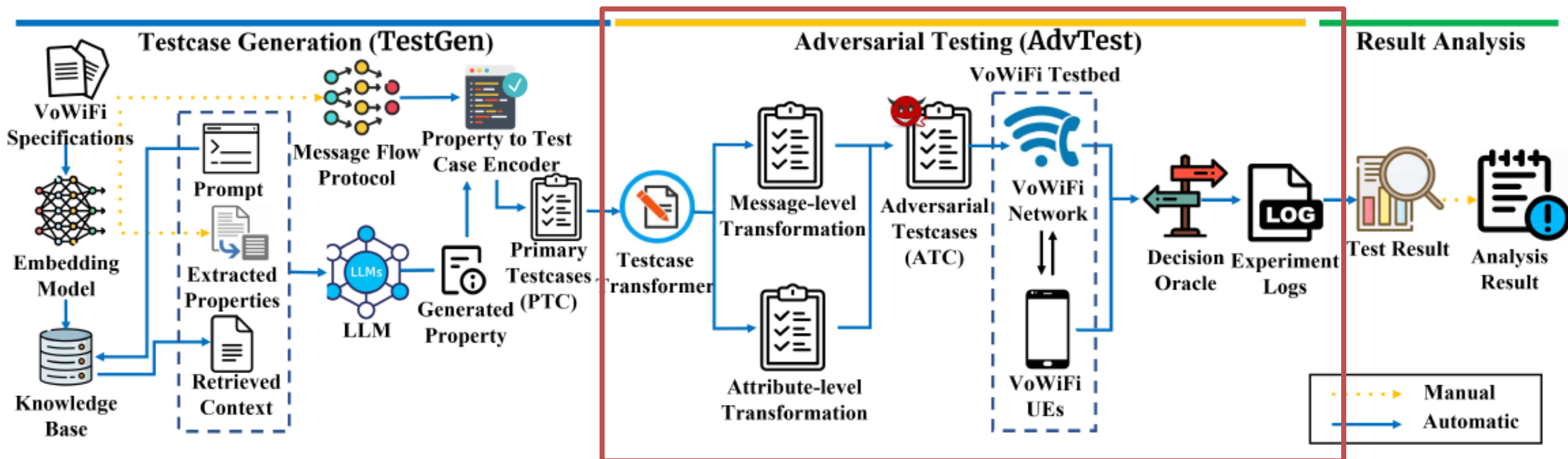
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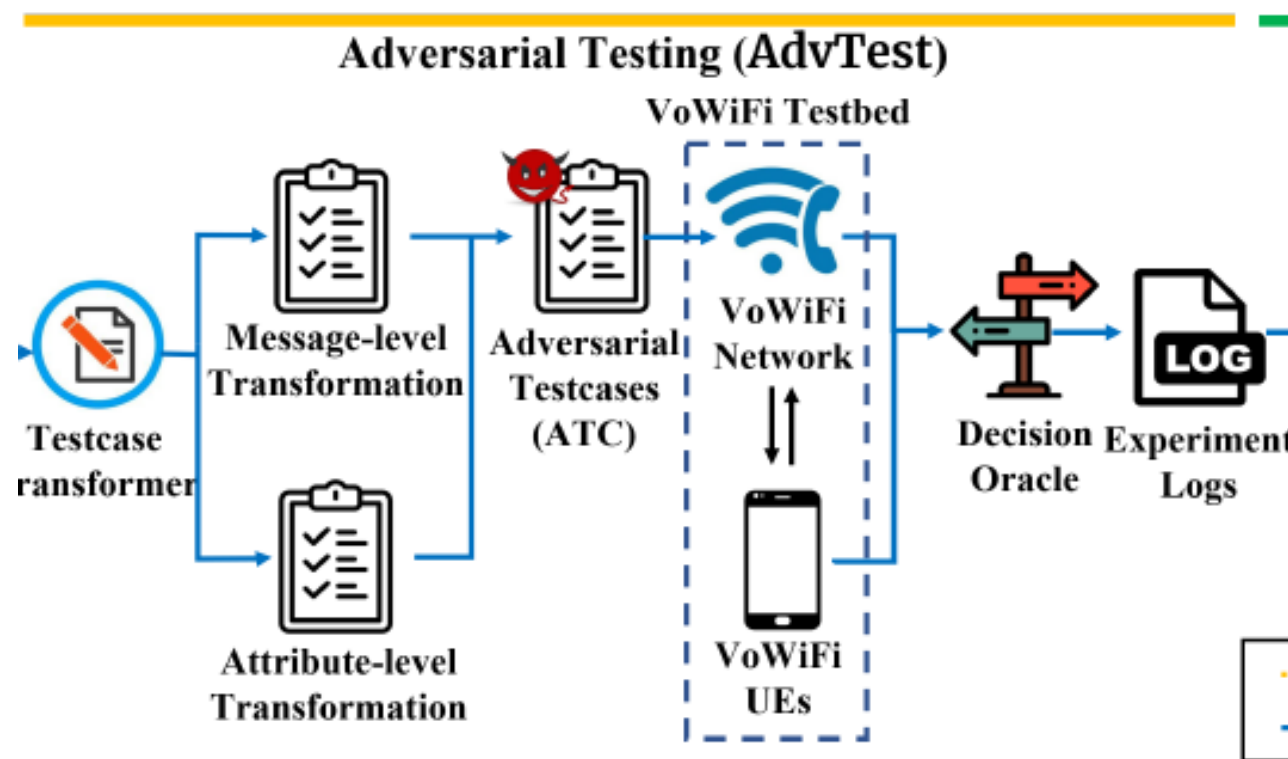
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AdvTest

It transforms PTCs into ATCs by initializing with actual values, runs ATCs, and detects misbehavior by decision oracles



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27        }
28      ]
29    }
30  ]
31 ]}

```

Message-level transformation

- 1) Message substitution
- 2) Message replay

Attribute-level transformation

- 1) Attribute value update
- 2) Attribute drop

AdvTest

It transforms PTCs into ATCs by initializing with actual values, runs ATCs, and **detects** misbehavior by decision oracles

Function oracle

It raises an alert if a UE responds to an adversarial message

Liveness oracle

It checks whether a UE can re-execute the protocol after the adversarial session



Evaluation on TestGen

RQ1. How faithful and relevant are the properties generated by TestGen?

* ROUTE: Recall-oriented understudy for gisting evaluation

Model	Generation Performance						Retrieval Performance	
	ROUTE*						Contextual Precision	Contextual Recall
	Precision		Recall		F1-score			
	RAG	No RAG	RAG	No RAG	RAG	No RAG		
Mistral-7B-v0.1	0.77	0.53	0.51	0.37	0.57	0.39	0.62	0.51
Llama-3-8B-Instruct	0.65	0.48	0.76	0.49	0.68	0.46	0.92	0.61
Llama-3.1-8B-Instruct	0.68	0.45	0.84	0.52	0.73	0.46	0.93	0.56

TestGen efficiently produces properties with RAG, showing much advanced models generates better properties

Result of AdvTest

RQ2. How effective is VWAttacker in finding?

No.	Testcase	Result
Support of Weak Algorithms in IKEv2		
1-5	A weak algorithm in IKE_SA_INIT	IKE_SA with a weak algorithm
Support of Weak Algorithms in SIP		
6-7	A weak algorithm in 401_unauthorized	A weak algorithm on SIP
Zero DH Key with IKE_SA_INIT		
8	An IKE_SA_INIT without KE_PAYLOAD	The DH shared secret is set to 0
Nonce Bypass with IKE_SA_INIT		
9	An IKE_SA_INIT without NONCE_PAYLOAD	The nonce value from ePDG is set to 0 on UE
DH Group Downgrade		
10	A weaker group in INVALID_KEY_PAYLOAD	The downgraded DH group

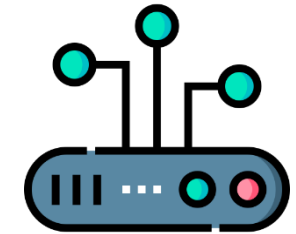
VWAttacker finds 10 issues after testing 1,164 ATCs on 21 UEs

Support of Weak Algorithms

```
{
  "testcase": [
    ...,
    {
      "name": "ike_sa_init_response",
      ...
      "sub": [
        {
          "name": "encryption_algorithm",
          "op": "update",
          "type": "uint16",
          "value": 12
        }
        ...
      ]
    }
  ]
}
```



UE



ePDG

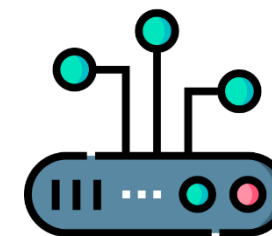
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```



UE

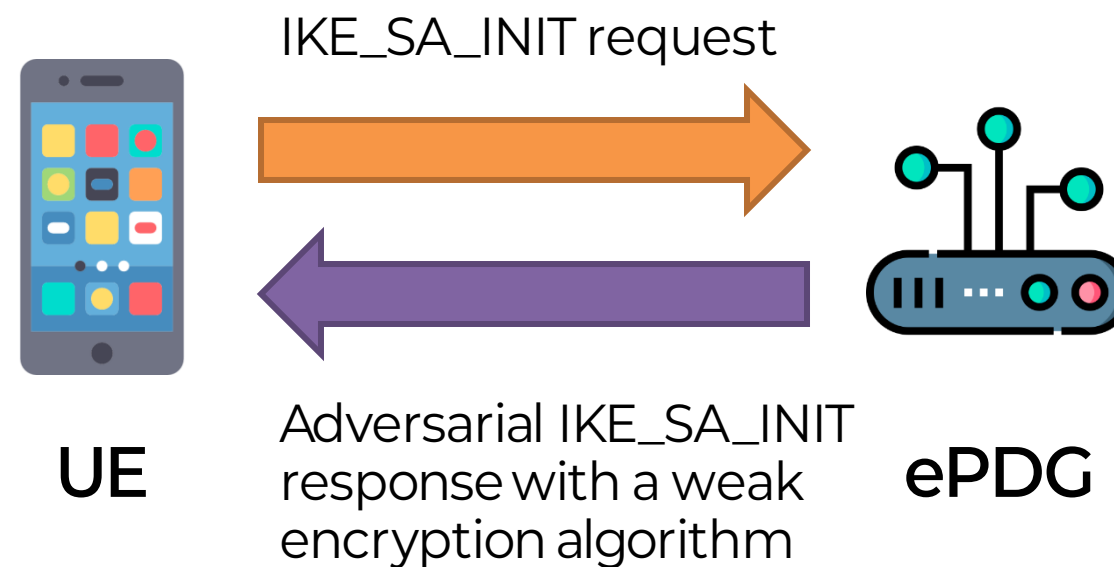
IKE_SA_INIT request



ePDG

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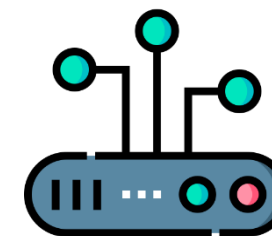
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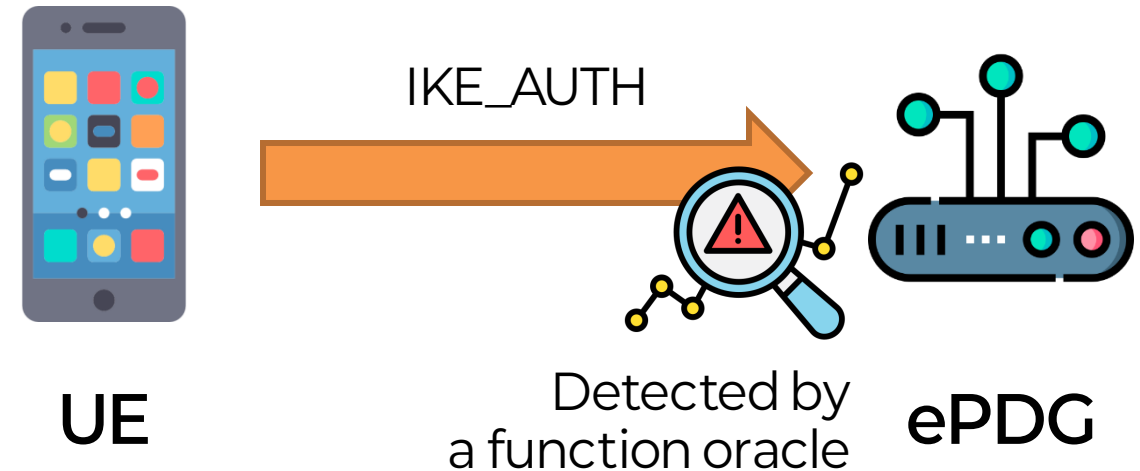
IKE_AUTH



ePDG

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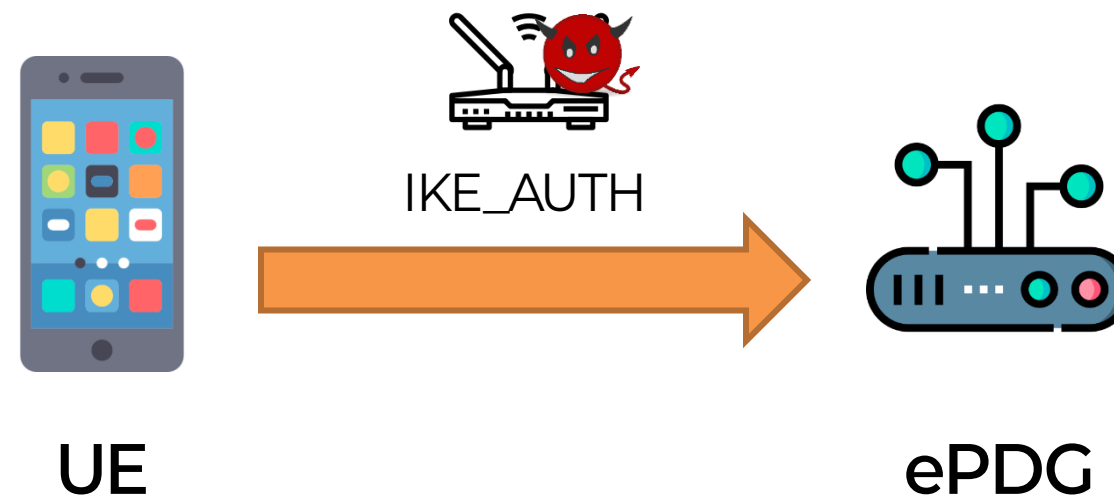


* This issue is acknowledged as CVE-2025-20667
(Inadequate encryption strength in Modem)

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An attacker in the middle of UE and ePDG can weaken channels for IKE_AUTH



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Summary

Motivation

No **systematic framework** for analyzing VoWiFi implementations **with** a **complete open-source VoWiFi UE testbed**

VWAttacker

The first **systematic** and **adversarial property-guided** framework to evaluate the security of VoWiFi UE implementations

- A VoWiFi testbed
- TestGen: Property extraction by LLM with RAG
- AdvTest: Adversarial testing with JSON-encoded adversarial testcases

Findings

10 issues on 21 UEs from 14 vendors
(Responsibly disclosed to related baseband vendors and others)

Thank you for listening!

Q&A

Source code:

<https://github.com/hw5773/vowifi-ue-testing-framework>